Functions & Transformations

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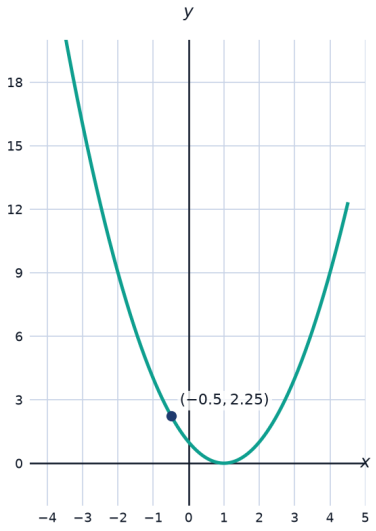
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Evaluating functions

If $f(x) = 3x - 4$, find $f(5)$.

2 [Foundational] • Target: Evaluating functions

If $f(x) = x^2 - 2x + 1$, find $f(-2)$.



3 [Foundational] • Target: Evaluating functions

If $g(x) = \sqrt{x + 5}$, find $g(4)$.

4 [Foundational] • Target: Evaluating functions

If $h(x) = \frac{2x}{x - 1}$, find $h(3)$.

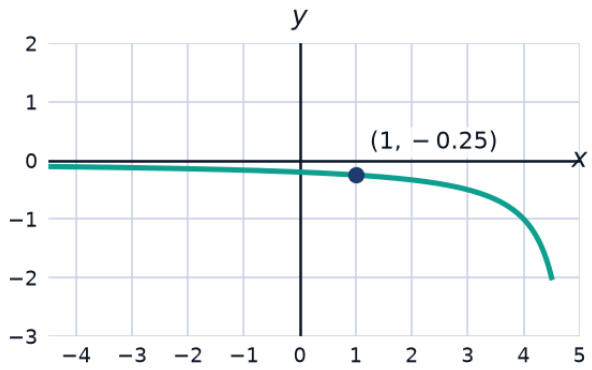
5 [Foundational] • Target: Domain and range

State the domain of $f(x) = \sqrt{x - 3}$.

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6 [Foundational] • Target: Domain and range

State the domain of $f(x) = \frac{1}{x-5}$.



7 [Foundational] • Target: Domain and range

State the domain of $f(x) = \sqrt{7-x}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Domain and range

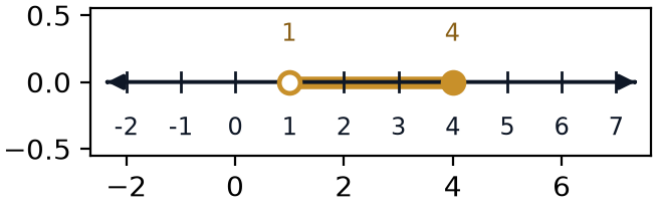
State the range of $f(x) = x^2 + 2$.

9 [Intermediate] • Target: Transformations: shifts

The graph of $y = x^2$ is shifted 3 units right and 2 units down. Write the new equation.

10 [Intermediate] • Target: Transformations: shifts

The graph of $y = |x|$ is shifted 4 units left and 1 unit up. Write the new equation.



11 [Intermediate] • Target: Transformations: shifts

Describe the transformation that takes $y = x^2$ to $y = (x+2)^2 - 5$.

12 [Intermediate] • Target: Transformations: reflections and stretches

The graph of $y = x^2$ is reflected over the x-axis and stretched vertically by a factor of 3. Write the new equation.

13 [Intermediate] • Target: Transformations: reflections and stretches

Describe the transformation that takes $y = \sqrt{x}$ to $y = 2\sqrt{x}$.

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14 [Intermediate] • Target: Transformations: reflections and stretches

Describe the transformation that takes $y = x^2$ to $y = -(x - 1)^2 + 4$.

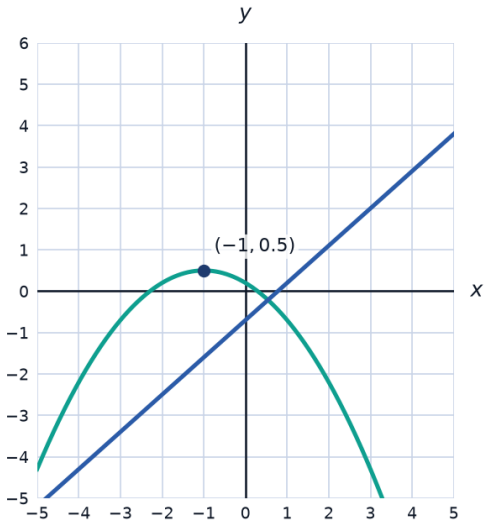
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Composite functions • Cross-Lesson: 9.01 Statistics & Probability

Let $f(x) = 2x + 1$ and $g(x) = x^2 - 3$. Find $(f \circ g)(2)$.

16 [Challenging] • Target: Composite functions • Cross-Lesson: 1.01 Algebra & Equations

Using the same f and g, find $(g \circ f)(2)$.



17 [Challenging] • Target: Composite functions • Cross-Lesson: 3.01 Sequences & Series

Using the same f and g, find a general formula for $(f \circ g)(x)$.

18 [Challenging] • Target: Composite functions • Cross-Lesson: 4.01 Trigonometry

Using the same f and g, find a general formula for $(g \circ f)(x)$.

19 [Challenging] • Target: Inverse functions • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Find the inverse of $f(x) = 3x - 2$.

20 [Challenging] • Target: Inverse functions • Cross-Lesson: 6.01 Limits & Derivatives

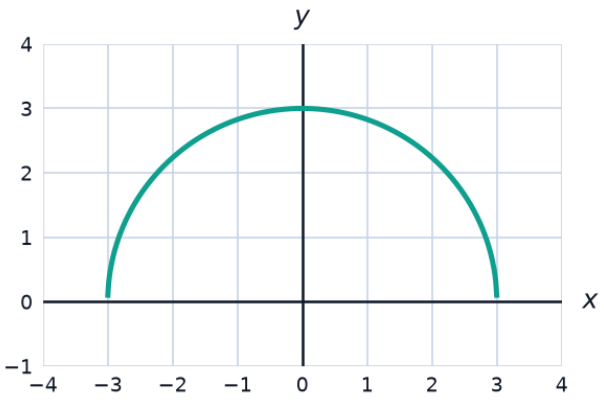
Find the inverse of $f(x) = \frac{x-4}{5}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Domain and range

The graph shows $f(x)$ defined only where plotted. State its domain.



a. $[-3, 3]$

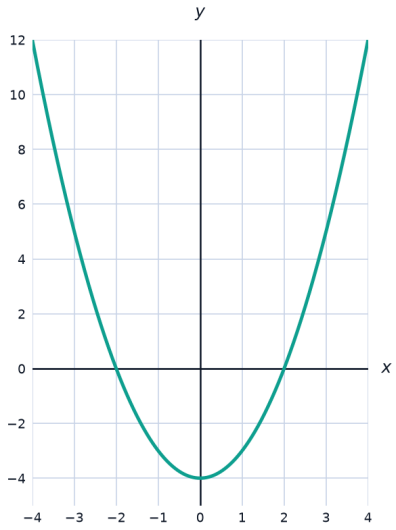
b. All real numbers

c. $[-4, 4]$

d. $[0, 3]$

22 [Foundational] • Target: Even and odd functions

The graph shown is symmetric about the y-axis. Is the function even, odd, or neither?



a. Odd

b. Neither

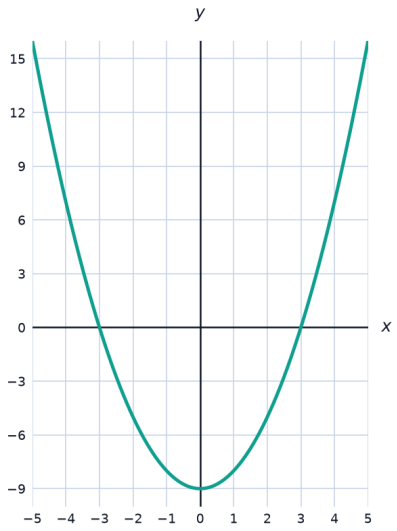
c. Cannot be determined

d. Even

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23 [Foundational] • Target: Reading a function graph

The graph shows $y = f(x) = x^2 - 9$. Find $f(3)$ by reading the graph.



a. -9

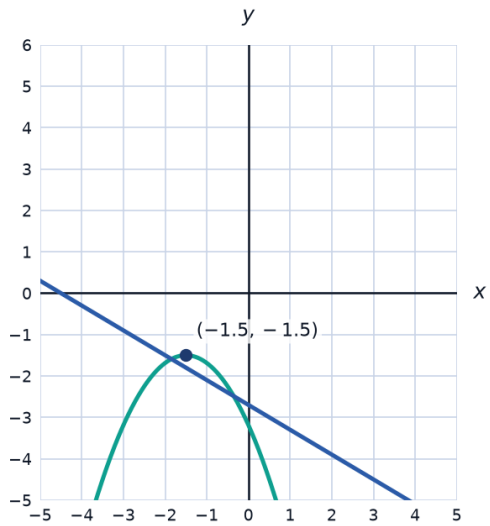
b. 6

c. 9

d. 0

24 [Foundational] • Target: Reading a function graph

Using the same graph, find the values of x where $f(x) = 0$ (the x-intercepts).



a. $x = 0$ and $x = 9$

b. $x = -9$ and $x = 9$

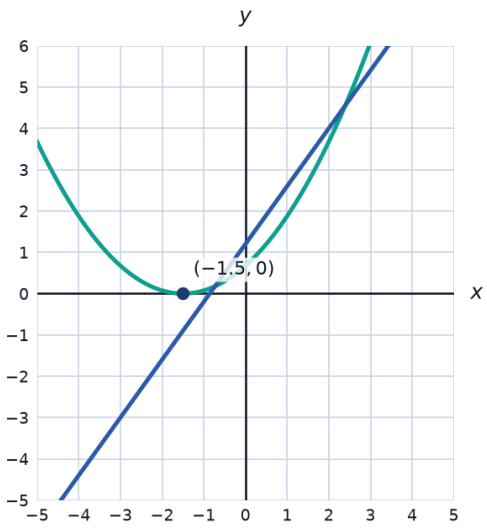
c. $x = 3$ only

d. $x = -3$ and $x = 3$

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25 [Foundational] • Target: Reading a function graph

Using the same graph, find the y-intercept.



a. -3

b. 9

c. 0

d. -9

26 [Foundational] • Target: Evaluating functions

If $f(x) = 2x^2 + 1$, find $f(3) - f(1)$.

a. 18

b. 8

c. 16

d. 22

27 [Foundational] • Target: Transformations: shifts

The graph of $y = |x|$ is shifted 2 units left and 4 units up. Write the new equation.

a. $y = |x + 4| + 2$

b. $y = |x + 2| + 4$

c. $y = |x + 2| - 4$

d. $y = |x - 2| + 4$

28 [Foundational] • Target: Transformations: shifts

Describe the transformation that takes $y = x^2$ to $y = (x - 3)^2 + 2$.

a. Shift right 2, up 3

b. Shift right 3, down 2

c. Shift left 3, up 2

d. Shift right 3, up 2

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Transformations: shifts**

The graph of $y = \sqrt{x}$ is shifted 6 units right. Write the new equation.

a. $y = \sqrt{x + 6}$

b. $y = \sqrt{x} - 6$

c. $y = \sqrt{x} + 6$

d. $y = \sqrt{x - 6}$

30 **[Intermediate]** • **Target: Transformations: reflections and stretches**

Describe the transformation that takes $y = x^2$ to $y = -(x + 2)^2 + 1$.

a. Reflect over x-axis, shift right 2, down 1

b. Reflect over y-axis, shift right 2, up 1

c. Reflect over x-axis, shift left 2, up 1

d. Shift left 2, up 1 (no reflection)

31 **[Intermediate]** • **Target: Transformations: reflections and stretches**

The graph of $y = \sqrt{x}$ is reflected over the y-axis. Write the new equation.

a. $y = -\sqrt{-x}$

b. $y = -\sqrt{x}$

c. $y = \sqrt{-x}$

d. $y = \sqrt{x}$

32 **[Intermediate]** • **Target: Composite functions**

Let $f(x) = x + 4$ and $g(x) = 2x$. Find $(f \circ g)(5)$.

a. 10

b. 9

c. 14

d. 18

33 **[Intermediate]** • **Target: Inverse functions**

Find the inverse of $f(x) = x^3 - 2$.

a. $f^{-1}(x) = \sqrt[3]{x} + 2$

b. $f^{-1}(x) = (x + 2)^3$

c. $f^{-1}(x) = \sqrt[3]{x - 2}$

d. $f^{-1}(x) = \sqrt[3]{x + 2}$

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34 [Intermediate] • Target: Inverse functions

Find the inverse of $f(x) = 5x + 10$.

a. $f^{-1}(x) = \frac{x-10}{5}$

b. $f^{-1}(x) = \frac{x+10}{5}$

c. $f^{-1}(x) = \frac{x}{5} - 10$

d. $f^{-1}(x) = 5x - 10$

35 [Intermediate] • Target: Piecewise functions

A function is defined by $f(x) = 2x$ for $x \leq 0$, and $f(x) = x^2 + 1$ for $x > 0$. Find $f(0) + f(3)$.

a. 1

b. 11

c. 0

d. 10

36 [Intermediate] • Target: Even and odd functions

Determine whether $f(x) = x^3 - 4x$ is even, odd, or neither.

a. Neither

b. Both even and odd

c. Even

d. Odd

37 [Intermediate] • Target: Even and odd functions

Determine whether $f(x) = x^2 + 3x$ is even, odd, or neither.

a. Both even and odd

b. Neither

c. Odd

d. Even

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Operations on functions • Cross-Lesson: 4.01 Trigonometry

Let $f(x) = 2x$ and $g(x) = x + 6$. Find $(f - g)(x)$.

a. $3x + 6$

b. $x - 6$

c. $x + 6$

d. $-x - 6$

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39 [Challenging] • Target: Word problems: functions in context • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

A company's daily profit (in dollars) from selling x units is modeled by $P(x) = -2x^2 + 80x - 300$. Find the profit when 15 units are sold.

a. 450 dollars

b. 150 dollars

c. 600 dollars

d. 900 dollars

40 [Challenging] • Target: Word problems: functions in context • Cross-Lesson: 6.01 Limits & Derivatives

The Fahrenheit conversion of a Celsius temperature is $F(C) = \frac{9C}{5} + 32$. Find $F(20)$.

a. 72°F

b. 68°F

c. 36°F

d. 52°F

41 [Challenging] • Target: Word problems: functions in context • Cross-Lesson: 7.01 Integration & Applications

A ball's height (in meters) t seconds after being thrown is $h(t) = -5t^2 + 20t + 2$. Find the height at $t = 2$ seconds.

a. 2

b. 18

c. 22

d. 42

42 [Challenging] • Target: Functions & Transformations • Cross-Lesson: 8.01 Exponentials & Logarithms

If $f(x) = x^2 - 6x - 7$, find $f(6)$.

a. -7

b. 7

c. -10

d. -4

43 [Challenging] • Target: Functions & Transformations • Cross-Lesson: 9.01 Statistics & Probability

If $f(x) = -3x^2 - 5x + 9$, find $f(-3)$.

a. 0

b. -6

c. -3

d. 3

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44 [Challenging] • Target: Functions & Transformations • Cross-Lesson: 1.01 Algebra & Equations

If $f(x) = -x^2 + 7x + 1$, find $f(-1)$.

a. -4

b. 7

c. -7

d. -10

45 [Challenging] • Target: Functions & Transformations • Cross-Lesson: 3.01 Sequences & Series

If $f(x) = -3x^2 - x - 1$, find $f(4)$.

a. -56

b. 53

c. -50

d. -53

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